

# 2015 RAA

## Pre-Calculus

Name ANSWERS

School \_\_\_\_\_

(Use full school name – no abbreviations)

\_\_\_\_\_ Correct X **2** pts. ea. =

**Note:** All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. <u>          <math>2\sqrt{2}</math>          </u> (Must be this exact answer.)</p>                                                                    | <p>11. <u>          22.02          </u> (Must be this decimal, "degrees" or ° optional.)</p>                                                                             |
| <p>2. <u>          (15,1)          </u> (Must be this ordered pair.)</p>                                                                                    | <p>12. <u>          -3          </u></p>                                                                                                                                 |
| <p>3. <u>          <math>7i, -7i</math> OR <math>0 + 7i, 0 - 7i</math>          </u> (Must have both answers in either form or order, but may use +/-.)</p> | <p>13. <u>          2390          </u></p>                                                                                                                               |
| <p>4. <u>          12          </u></p>                                                                                                                     | <p>14. <u>          62          </u></p>                                                                                                                                 |
| <p>5. <u>          <math>\frac{3}{2}</math>          </u> (Must be this reduced improper fraction.)</p>                                                     | <p>15. <u>          <math>\frac{2}{7}</math>          </u> (Must be this reduced common fraction.)</p>                                                                   |
| <p>6. <u>          <math>\frac{1}{8}</math>          </u> (Must be this reduced common fraction.)</p>                                                       | <p>16. <u>          <math>70.57^\circ</math>          </u> (Must be this decimal, "degrees" or ° optional.)</p>                                                          |
| <p>7. <u>          123          </u></p>                                                                                                                    | <p>17. <u>          <math>(4, \sqrt{3})</math>          </u> (Must be this exact ordered pair.)</p>                                                                      |
| <p>8. <u>          2          </u> (This value only.)</p>                                                                                                   | <p>18. <u>          502          </u> ("Zeros optional.)</p>                                                                                                             |
| <p>9. <u>          55.49          </u> (Must be this decimal, "degrees" or ° optional.)</p>                                                                 | <p>19. <u>          <math>-\frac{\sqrt{7}}{4}</math> OR <math>-\frac{1}{4}\sqrt{7}</math> OR <math>-0.25\sqrt{7}</math>          </u> (Or exact reduced equivalent.)</p> |
| <p>10. <u>          <math>\frac{31}{35}</math>          </u> (Must be this reduced common fraction.)</p>                                                    | <p>20. <u>          10          </u></p>                                                                                                                                 |